



SHRI RAMSWAROOP MEMORIAL COLLEGE OF ENGINEERING & MANAGEMENT

Ref. No. EE/PAC/2024-25/01

4th October 2024

To,
Director
SRMCEM, Lucknow
Subject: PAC Meeting Electrical Engineering Department

Respected Sir,
The PAC Meeting of Electrical Engineering Department has been proposed on 5th October, 2024 from 10:00 AM onwards. The following PAC members will attend the meeting.

Venue: EE/EC Department (C-309)

S. No.	Name	Post and Affiliation	Responsibility
1.	Prof. (Dr.) Indu Prabha Singh	HOD (EE/ EC) SRMCEM, Lucknow	Convenor
2.	Er. Jayati Vaish	AP, EE Dept. SRMCEM, Lucknow	Member Secretary
3.	Er. Amit Kumar	AP EE Dept. SRMCEM, Lucknow	Member
4.	Dr. Anshul Agarwal	Associate Professor, EE Dept., BBAU, Lucknow	External Expert
5.	Prof. K. S. Singh	SE(R), UPPCL	Industry Expert
6.	Dr. Vijay Tiwari	Assistant Professor, REC, Kannauj	Alumni 2006
7.	Er. Naveen Pandey	Assistant Consultant, TCS	Alumni 2009

It is for your kind information and also for kindly approval.

Prof. (Dr.) Indu Prabha Singh
HoD, EE&EN/EC Department
SRMCEM, Lucknow

Prof. (Dr.) Bhavesh Kumar Chauhan
Director
SRMCEM, Lucknow



SHRI RAMSWAROOP MEMORIAL COLLEGE OF ENGINEERING & MANAGEMENT

Agenda of 17th Program Assessment Committee (PAC) Odd Semester 2024-25

Curriculum Gaps for the 2024-25 (ODD) Semester

Agenda 1: 3rd Semester Subjects & practical

- 1.1** Design of PCB, power supply, step up/step down transformer (practical approach) should be included.
- 1.2** BEE-301 EMFT subject should be in 4th or 5th semester. It becomes lengthy for 3rd semester.
- 1.3** Circuit simulation Lab (BEE-351) includes experiments related to network. Therefore, this lab should be included in 4th semester with network analysis and synthesis.

Agenda 2: 5th Semester Subjects and practical

- 2.1** Unit 4: Frequency response analysis in BEE-502 (Control System) should be divided in two parts. In unit 5 state space analysis can be excluded.
- 2.2** Unit 3 in BEE-052 (Sensors and Transducers) is Machine Vision is quite different from rest of the units. It can further be modified with some smart sensors.

Agenda 3: 7th Semester Subjects and practical

- 3.1** Core subjects must be included as compulsory subjects (e.g., Power System Protection).
- 3.2** Certified Training program should be included in curriculum in final year.

Prof. (Dr.) Indu Prabha Singh
HoD, EE & EN/ EC
SRMCEM, LUCKNOW



SHRI RAMSWAROOP MEMORIAL COLLEGE OF ENGINEERING & MANAGEMENT

Ref. No. EE/PAC/2024-25/02

5th October 2024

Minutes of Meeting

The Program Assessment Committee (PAC)/17th PAC meeting is held in the Electrical Engineering department. The details of meeting are-

Date of Meeting	Saturday, 5th October 2024	Timing	10:00 pm– 12:00 pm
Chaired By	Dr. Indu Prabha Singh, HoD	Co-ordinated By	Er. Jayati Vaish
Attendees	PAC Committee members		

1. The following members were present in the department.

S. No.	Name	Post and Affiliation	Responsibility
1.	Prof. (Dr.) Indu Prabha Singh	HOD (EE & EN/ EC) SRMCEM, Lucknow	Convenor
2.	Er. Jayati Vaish	AP, EE Dept. SRMCEM, Lucknow	Member Secretary
3.	Dr. Anshul Agarwal	Associate Professor, EE Dept., BBAU, Lucknow	External Expert
4.	Dr. Vijay Kumar Tiwari	Assistant Professor, EE Dept, REC Kannauj	Alumni 2006
5.	Er. Naveen Pandey	Consultant, TCS	Alumni 2009

2. The Convener, PAC meeting welcomed and introduced external members of Program Assessment Committee and thanked him for sparing his valuable time to attend the meeting.

3. The Member Secretary, PAC meeting with the permission of the Convener, presented the agenda of PAC.

5. The PAC of the department had discussed following agendas:

Agenda 1: 3rd Semester Subjects & practical

1.1 Design of PCB, power supply, step up/step down transformer (practical approach) should be included.

Suggestions: These design and fabrication experiments can be included as extra practical in Electrical Workshop Lab or in their respective course labs.

1.2 BEE-301: EMFT subject should be in 4th or 5th semester. It becomes tough for 3rd semester.

1.3 Circuit simulation Lab (BEE-351) includes experiments related to network analysis subject. Therefore, this lab should be included in 4th semester with network analysis and synthesis.

Suggestions: The change in practical or lab can be proposed to AKTU through mail for revision in syllabus. Since in 4th semester already network lab is there, so change in experiments can be proposed to AKTU.

Agenda 2: 5th Semester Subjects and practical

2.1 Unit 4: Frequency response analysis in BEE-502 (Control System) should be divided in two parts. In unit 5 state space analysis can be excluded.

Suggestions: The study of state space analysis for fuzzy logic is required. However, in UNIT 3, the introduction of frequency response can be included. The revision in syllabus can be communicated to the AKTU through mail.

2.2 Unit 3 in BEE-052 (Sensors and Transducers) is Machine Vision which is quite different from rest of the units. It can further be modified with some smart sensors.

Suggestions: The syllabus itself gives the detail analysis of sensors. So, requirement of modification is not required.

Agenda 3: 7th Semester Subjects and practical

3.1 Core subjects must be included as compulsory subjects (e.g., Power System Protection).

Suggestions: The addition of subject as core can be proposed to AKTU through mail for revision in syllabus. Further the students are also motivated to choose it as core subject.

3.2 Certified Training program should be included in curriculum in final year.

Suggestion: The students cover the internship and summer training in this semester.



Prof. (Dr.) Indu Prabha Singh
Convenor

Dr. Anshul Agarwal
External Expert



Er. Jayati Vaish
Member Secretary



Shri Ramswaroop Memorial College of Engineering & Management

Department of Electrical Engineering

Ref. No. EN & EE /PAC/2024-25/03

Date: 10th Oct, 2024

Action Taken Report for Minutes of 15th PAC Meeting

As per PAC committee suggestions, following action has been taken place:

1. **For Point 1.1:** All project guides and instructors have been requested to include study/experiments on PCB design for projects. Further, they will motivate students to make minor projects by using PCB Lab.
2. **For Point 1.2:** The concerned faculty member has been advised to do some more efforts in the EMFT subject and give them practical examples.
3. **For Point 1.3:** The proposed change in experiments of BEE-351 (Circuit Simulation Lab) is mailed to AKTU.
4. **For Point 2.1:** The proposed change in experiments of BEE-502 (Control System) is mailed to AKTU
5. **For Point 2.2:** The faculty has been advised to give an overview of smart sensors.
6. **For Point 3.1:** It has been advised to faculty members to guide students to understand the need of core subjects so that they choose core subjects in Elective. The mail is also sent to AKTU to include KEE077: Power System Protection as core subject in 7th semester.
8. **For Point 3.2:** The faculties are advised to motivate students to do NPTEL course related to Start-up/Entrepreneurship and innovation. This will help the students to inculcate in other training programs beneficial for their career. Industrial Visit , Guest Lectures have been arranged to fulfil the Gap between Industry and academia.

Prof. (Dr.) Indu Prabha Singh
Head of Department
EE/ EC
SRMCEM, LKO

BEE 351- CIRCUIT SIMULATION LAB

Pre-requisites of course: Basic Electrical Engineering

Course Outcome		Knowledge Level, KL
Upon the completion of the course, the student will be able to:		
CO 1	Apply knowledge of MATLAB in various simulation circuits.	K2
CO2	Apply the knowledge of basic circuit law, nodal and mesh analysis for given circuit.	K3
CO3	Analysis of the AC and DC circuits using simulation techniques.	K3
CO4	Analysis of transient response of AC circuits.	K2
CO5	Analysis of single and three phase circuits using various loads.	K2

KL- Bloom's Knowledge Level (K1 , K2 , K3 , K4 , K5 , K6)

K1– Remember K2– Understand K3– Apply K4– Analyze K5– Evaluate K6– Create

List of Experiments

Ten experiments to be performed

1. Introduction to MATLAB and its basic commands
2. Determine the root of a polynomial
3. Determination of polynomial using method for least square curve fitting
4. Solution of differential equation using 4th order runge - kutta method
5. Determination of time response of an RLC circuit
6. Single line Modeling of DC motor
7. Step, Ramp and impulse response of transfer function
8. Generation of single and three phase sinusoidal waveform
9. PWM based waveform generation
10. Single phase uncontrolled half wave rectifier using R and RL load
11. Single phase uncontrolled full wave rectifier using R and RL load
12. Three phase uncontrolled full wave rectifier using R and RL load

Note: Minimum ten experiments are to be performed from the following list:

PROPOSED SYLLABUS BY AKTU

BEE 502: CONTROL SYSTEM

Pre-requisites of course: Basic signal systems

Course Outcome		Knowledge Level, KL
Upon the completion of the course, the student will be able to:		
CO 1	Obtain transfer functions to predict the correct operation of open loop and closed loop control systems and identify the basic elements, structures and the characteristics of feedback control systems.	K3
CO2	Measure and evaluate the performance of basic control systems in time domain. Design specification for different control action.	K4
CO3	Analyze the stability of linear time-invariant systems in time domain using Routh- Hurwitz criterion and root locus technique. Analyze the system in frequency domain and construct the polar plots.	K4
CO4	Determine the stability of linear time-invariant systems in frequency domain using Nyquist criterion and Bode plot.	K4
CO5	Design different type of compensators to achieve the desired performance of control System by root locus and Bode plot method. Develop and analyze the intermediate states of the system using state space analysis.	K4

KL- Bloom's Knowledge Level (K1, K2, K3, K4, K5, K6)

K1 – Remember K2 – Understand K3 – Apply K4 – Analyze K5 – Evaluate K6 – Create

Detailed Syllabus:

Unit-I:

Control System Concepts: Elements of control systems, concept of open loop and closed loop systems, Examples and application of open loop and closed loop systems, Mathematical Modelling of Physical Systems (Electro Mechanical), Determination of transfer function by block diagram reduction techniques and signal flow method using Mason's gain formula, Basic Characteristics of negative feedback control systems.

Control System Components: Constructional and working concept of AC & DC servomotor, synchro's, stepper motor and tachometer.

Unit-II:

Time Response Analysis: Standard test signals, time response analysis of first and second order systems, time response specifications of second order system for unit step input, location of roots of characteristics equation and corresponding time response, steady state errors and error constants.

Basic modes of feedback control: Proportional, Derivative, Integral and PID controllers, controller design using Ziegler Nichols.

Unit-III:

Stability and Algebraic Criteria: Concept of stability and its necessary conditions, Routh-Hurwitz criteria and its limitations.

Root Locus Technique: Salient features of root locus plot, Procedure for plotting root locus, root contours.

Frequency Response Analysis: Frequency Response analysis from transfer function model, Construction of polar and inverse polar plots.

Unit-IV:

Stability in Frequency Domain: Nyquist stability criterion, Determination of gain and phase margin from Bode & Nyquist Plots, Correlation between time and Frequency Responses.

Unit-V

Introduction to Design: The design problems and preliminary considerations of lead, lag and lead-lag compensation networks, design of closed loop systems using compensation techniques in time and frequency domains.

State Space Technique: Concept of single input and multiple input and multiple output system, State-space model of physical system, conversion of state-space to transfer function model and vice-versa, State transition matrix, Concept of controllability and observability and their testing.

Text Book:

1. I. J. Nagrath & M. Gopal, "Control System Engineering", 6th Ed. New Age International Publishers, 4th Edition 2018
2. M. Gopal, "Control System Principles and Design", McGraw Hill 4th Edition
3. Ogata, "Modern Control Engineering, 5th Edition", Pearson Education, 2015
4. B.C. Kuo & Farid Golnaraghi, "Automatic Control Systems", 10th Edition, McGraw Hill
5. D. Roy Choudhary, "Modern Control Engineering", Prentice Hall of India.
6. A.K. Jairath, "Problems and solutions of Control System with Essential Theory", CBS Publishers

Reference Books:

1. (Schaums Outlines Series) Joseph J. Distefano III, Allen R. Stubberud, Ivan J. Williams, "Control Systems", 3rd Edition, McGraw Hill, Special Indian Edition, 2010.
2. Norman S. Mise, Control System Engineering, Wiley Publishing Co.
3. Ajit K Mandal, "Introduction to Control Engineering" New Age International.
4. R.T. Stefani, B.Shahian, C.J.Savant and G.H. Hostetter, "Design of Feedback Control Systems" Oxford University Press.
5. Samarjit Ghosh, "Control Systems theory and Applications", Pearson Education.

BEE 351- CIRCUIT SIMULATION LAB

Pre-requisites of course: Basic Electrical Engineering

Course Outcome		Knowledge Level, KL
Upon the completion of the course, the student will be able to:		
CO 1	Apply the knowledge of basic circuit law, nodal and mesh analysis for given circuit.	K2
CO2	Analysis of the AC and DC circuits using simulation techniques.	K3
CO3	Analysis of transient response of AC circuits.	K3
CO4	Evaluation and analysis of two-port network parameters.	K2
CO5	Estimation of parameters of different filters.	K2

KL- Bloom's Knowledge Level (K1 , K2 , K3 , K4 , K5 , K6)

K1– Remember K2– Understand K3– Apply K4– Analyze K5– Evaluate K6– Create

List of Experiments

Ten experiments to be performed

1. Create Verification of principle of Superposition with AC sources using Multisim/ PSPICE.
2. Verification of Thevenin and Maximum Power Transfer theorems in AC Circuits using Multisim/ PSPICE.
3. Verification of Norton theorems in AC Circuits using Multisim/ PSPICE.
4. Verification of Tellegen's theorem for two networks of the same topology using Multisim/ PSPICE.
5. Determination of Z and h-parameters (DC only) for a network and computation of Y and ABCD Parameters using Multisim/ PSPICE.
6. Determination of driving point and transfer functions of a two-port ladder network and verify with theoretical values using Multisim/ PSPICE.
7. Determination of transient response of current in RL and RC circuits with step voltage input.
8. Determination of transient response of current in RLC circuit with step voltage input for under damped, critically damped and over damped cases.
9. Determination of image impedance and characteristic impedance of T and Π networks, using O.C. and S.C. tests.
10. Verification of parameter properties in inter-connected two port networks: series, parallel and cascade using Multisim/ PSPICE.
11. Determination of frequency response of a Twin – T-notch filter.
12. To determine attenuation characteristics of a low pass / high pass active filters.

Note: Any two experiments from above list should also be performed by students on Virtual Lab

PROPOSED SYLLABUS BY AKTU

BEE 502: CONTROL SYSTEM

Pre-requisites of course: Basic signal systems

Course Outcome		Knowledge Level, KL
Upon the completion of the course, the student will be able to:		
CO 1	Obtain transfer functions to predict the correct operation of open loop and closed loop control systems and identify the basic elements, structures and the characteristics of feedback control systems.	K3
CO2	Measure and evaluate the performance of basic control systems in time domain. Design specification for different control action.	K4
CO3	Analyze the stability of linear time-invariant systems in time domain using Routh- Hurwitz criterion and root locus technique.	K4
CO4	Determine the stability of linear time-invariant systems in frequency domain using Nyquist criterion and Bode plot.	K4
CO5	Design different type of compensators to achieve the desired performance of control System by root locus and Bode plot method. Develop and analyze the intermediate states of the system using state space analysis.	K4

KL- Bloom's Knowledge Level (K1, K2, K3, K4, K5, K6)

K1 – Remember K2 – Understand K3 – Apply K4 – Analyze K5 – Evaluate K6 – Create

Detailed Syllabus:

Unit-I:

Control System Concepts: Elements of control systems, concept of open loop and closed loop systems, Examples and application of open loop and closed loop systems, Mathematical Modelling of Physical Systems (Electro Mechanical), Determination of transfer function by block diagram reduction techniques and signal flow method using Mason's gain formula, Basic Characteristics of negative feedback control systems.

Control System Components: Constructional and working concept of AC & DC servomotor, synchro's, stepper motor and tachometer.

Unit-II:

Time Response Analysis: Standard test signals, time response analysis of first and second order systems, time response specifications of second order system for unit step input, location of roots of characteristics equation and corresponding time response, steady state errors and error constants.

Basic modes of feedback control: Proportional, Derivative, Integral and PID controllers, controller design using Ziegler Nichols.

Unit-III:

Stability and Algebraic Criteria: Concept of stability and its necessary conditions, Routh-Hurwitz criteria and its limitations. Root Locus Technique: Salient features of root locus plot, Procedure for plotting root locus, root contours.

Unit-IV:

Frequency Response Analysis: Frequency Response analysis from transfer function model, Construction of polar and inverse polar plots.

Stability in Frequency Domain: Nyquist stability criterion, Determination of gain and phase margin from Bode & Nyquist Plots, Correlation between time and Frequency Responses.

Unit-V

Introduction to Design: The design problems and preliminary considerations of lead, lag and lead-lag compensation networks, design of closed loop systems using compensation techniques in time and frequency domains.

State Space Technique: Concept of single input and multiple input and multiple output system, State-space model of physical system, conversion of state-space to transfer function model and vice-versa, State transition matrix, Concept of controllability and observability and their testing.

Text Book:

1. I. J. Nagrath & M. Gopal, "Control System Engineering", 6th Ed. New Age International Publishers, 4th Edition 2018
2. M. Gopal, "Control System Principles and Design", McGraw Hill 4th Edition
3. Ogata, "Modern Control Engineering, 5th Edition", Pearson Education, 2015
4. B.C. Kuo & Farid Golnaraghi, "Automatic Control Systems", 10th Edition, McGraw Hill
5. D. Roy Choudhary, "Modern Control Engineering", Prentice Hall of India.
6. A.K. Jairath, "Problems and solutions of Control System with Essential Theory", CBS Publishers

Reference Books:

1. (Schaums Outlines Series) Joseph J. Distefano III, Allen R. Stubberud, Ivan J. Williams, "Control Systems", 3rd Edition, McGraw Hill, Special Indian Edition, 2010.
2. Norman S. Mise, Control System Engineering, Wiley Publishing Co.
3. Ajit K Mandal, "Introduction to Control Engineering" New Age International.
4. R.T. Stefani, B.Shahian, C.J.Savant and G.H. Hostetter, "Design of Feedback Control Systems" Oxford University Press.
5. Samarjit Ghosh, "Control Systems theory and Applications", Pearson Education.